INCH-POUND

MIL-L-3122G
29 March 1991
SUPERSEDING
MIL-L-3122F
5 August 1971

MILITARY SPECIFICATION

LEATHER, CATTLEHIDE, FOR FOOTWEAR

UPPERS, CHROME TANNED, FATLIQUORED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers smooth finish, full grain and corrected grain, chrome tanned, fatliquored, cattlehide leather for footwear uppers.

1.2 Classification. The leather shall be of one type in the following treatments as specified (see 6.2).

Treatment A - Treated for resistance to water absorption
Treatment B - Treated for resistance to water penetration
Treatment C - Not treated for resistance to water

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5019 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8330

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-P-43432 - Packaging of Leather

STANDARDS

FEDERAL

FED-STD-311 - Leather, Methods of Sampling and Testing

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

Chromatic Transference Scale

(Copies may be obtained from the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 22709.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Standard sample. The leather, when embossed, shall match the standard sample for fine hair cell pattern and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.4).

3.3 Material. The leather shall be full grain or corrected grain, chrome tanned cattlehide sides.

3.4 Trim (form) and area. The leather sides shall be trimmed to conform with the definition of a "side" as specified in FED-STD-311. The area of the side shall be not more than 28 square feet. The area expressed in square feet shall be legibly marked in the butt area on the flesh side of the leather.

3.5 Thickness. The leather shall be not less than 4-1/2 ounces in the lightest area and not more than 6 ounces in the heaviest area when examined as specified in 4.4.4. The side shall be not less than 5 ounces in thickness when measured at a point 6 inches in from the backbone and 9 inches from the root of the tail.

NOTE: 1 ounce = 1/64 inch.

3.6 Color. Unless otherwise specified, the color of the grain side of the leather shall be black (see 6.2). The color of the flesh side of the leather shall be natural or black. When the flesh side is black, the color shall be produced by drum dyeing.

3.6.1 Staining. Except as noted below, the flesh side of the leather that has been drum dyed shall be tested for staining in accordance with 4.4.4. The stain on the test pad shall have an expanded AATCC value of not less than 3.

NOTE: Leather which has a natural colored flesh side that has been subjected to small amounts of dye added to the drum in order to level the shade of the natural color need not to be tested for staining.

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3.7 Finish. Corrected grain leather shall be buffed or snuffed only to the extent that is necessary to remove superficial blemishes. The grain side of the leather shall have a light application of finish containing only sufficient pigment to assist in making the color uniform. The finish of the grain side shall be smooth. A fine hair cell embossed pattern may be applied to the grain surface of the side. Printed leather shall not be permitted. If embossed, the leather shall match the standard sample for fine hair cell pattern. Leather previously dyed or finished in a color not specified and then redyed or refinished to match a specified color shall not be acceptable.

3.8 Paranitrophenol. Unless othewise specified (see 6.2), the leather shall contain not less than 0.18 percent and not more than 0.70 percent paranitrophenol fungicide when tested as specified in 4.4.4.

3.9 Water resistant treatment. When treatment A or B is specified, the leather shall be treated with a water resistant compound (see 6.7), and shall meet the following requirements as applicable.

3.9.1 Treatment A (water absorption). When treatment A is specified, specimens of the treated leather shall gain not more than 30 percent of their weight by water absorption when tested as specified in 4.4.4.

3.9.2 Treatment B (water penetration). When treatment B is specified, at least 80 percent of the specimens tested shall withstand not less than 15,000 cycles by method 8021 or not less than 4,000 cycles by method 8131 without water penetration, when tested as specified in 4.4.4.

3.10 Tearing strength (slot tear). At least 80 percent of the specimens tested shall have slot tearing values of not less than 35 pounds when tested as specified in 4.4.4.

3.11 Shrinkage. The leather shall not shrink before the temperature reaches 98°C ± 0.5°C for a period of 30 seconds when tested as specified in 4.4.4.

3.12 Stiffness. At least 80 percent of the specimens tested shall have a load scale reading of not more than 50 when tested as specified in 4.4.4.

3.13 Lubrication. The leather shall be processed with lubricating materials containing less than 10 percent hydrocarbon mineral oil, based on the moisture-free weight of the lubricating materials (see 6.6).

3.14 Chemical requirements. The leather shall conform to the chemical requirements in table I when tested as specified in 4.4.4.

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TABLE I. Chemical requirements

Characteristics	Minimum	Maximum
Chloroform soluble material, percent 1/	6.0	16.0
Chromic oxide, hide substance basis, percent	4.0	7.5
Non-determined material, percent	-	20.0
pH value	3.0	4.0
Total ash, percent 1/	-	10.0

1/ Calculated on a moisture-free basis.

3.15 Workmanship. The finished leather shall conform to the quality of product established by this specification, and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.2 and 4.4.3 and tested as specified in 4.4.4.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with the referenced documents.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 End item visual examination. The end items shall be examined for the defects listed below. The lot size shall be expressed in units of sides. The sample unit shall be one side. Any sample unit containing one or more defects shall be classified as a defective unit. Sampling for this examination shall be performed in accordance with FED-STD-311.

Examine	Defect
Trim	Not as specified
Color	Not as specified Color not uniform
Finish	Not finished on grain side Finish not smooth Embossing other than a fine hair cell pattern Printed design on leather Does not match standard sample for fine hair cell pattern Finish streaky, chipped or flaky Redyed or refinished to match specified color

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Quality of leather	Not clean; stain, discoloration or foreign matter Flesh side contains loose flesh exceeding 16 square inches in total area Hard, boney, boardy or loose pipey leather Any cut, hole, brand, hip mark, scratch, heavy vein, deep fat wrinkle and surface imperfections which individually or collectively impair the cutting value of more than 12 percent of the total area of a side
Marking	Omitted, illegible, or not in area specified.

4.4.3 End item dimensional examination. The end items shall be examined for the defects listed below. The lot size shall be expressed in units of sides. The sample unit shall be one side. Any sample unit containing one or more defects shall be classified as a defective unit. Sampling for these examinations shall be performed in accordance with FED-STD-311, except that a sample size of 15 and a rejection number of 2 shall be applicable for thickness determinations on lots consisting of 51 or more sample units.

Examine	Defect
Area (expressed in in terms of square feet)	Not as marked on the side Area greater than 28 square feet
Thickness 1/	Not as specified

1/ Thickness shall be determined in accordance with Method 1011 of FED-STD-311, except that no individual thickness measurement shall be less than the minimum or greater than the maximum specified.

4.4.4 End item testing. The methods of testing specified in FED-STD-311, wherever applicable and as listed in table II, shall be followed. Sampling procedure and location from which the sample unit is to be obtained shall be in accordance with FED-STD-311. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be rejected if any one of the following conditions exists:

a. More than three specimens fail requirements for stiffness.
b. More than three specimens fail requirements for water penetration or slot tearing strength.
c. Any specimen fails requirements for water absorption, shrinkage temperature or staining.
d. Any composite test result fails to meet the specified requirements.

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TABLE II. End item tests

Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Composite
Material identification	3.3	1/	-	-
Tannage	3.3	1/	-	-
Staining (drum dyed leather)	3.6.1	3021 2/	X	-
Paranitrophenol	3.8	6711	-	X
Water absorption (treatment A)	3.9.1	8112	X	-
Water penetration (treatment B)	3.9.2	8021 or 8131 3/	X	-
Tearing strength (slot tear)	3.10	2161	X	-
Shrinkage	3.11	7011	X	-
Stiffness	3.12	4.5.1	X	-
Lubrication	3.13	1/	-	-
Chloroform-soluble material	3.14	6341	-	X
Chromic oxide	3.14	6515	-	X
Non-determined material	3.14	4.5.2	-	X
pH value	3.14	6621	-	X
Total ash	3.14	6341 4/	-	X

1/ A certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ Except that the flesh side of the leather shall be in contact with the color transfer pad.

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3/ The test method used shall be referenced in the test report.

4/ The procedure for determining water soluble material shall be omitted. The ash shall be determined on the chloroform extracted specimen.

4.4.5 Packaging inspection. The sampling and inspection of the preservation, packaging, palletization, and marking shall be in accordance with the requirements of MIL-P-43432.

4.5 Methods of inspection.

4.5.1 Stiffness. The specimen shall be 2 by 1 inches cut with the long dimension perpendicular to the backbone. The stiffness of the specimen shall be determined as specified in Method 4211 of FED-STD-311, using a 1/2 inch span and a total moment weight equivalent to 0.5 inch-pound. The specimen shall be clamped firmly in the vise and the apparatus engaged until the specimen has been subjected to an angular deflection of 40 degrees. The apparatus shall be immediately readjusted by hand so as to completely remove the load on the specimen. Then the specimen shall be subjected to an angular deflection of 20 degrees following the procedure specified in Method 4211. The stiffness shall be recorded as the load scale reading.

4.5.2 Non-determined material. The percent non-determined material shall be calculated as follows:

Non-determined material, percent = 100 - (A + B + C)

Where: A = Percent chloroform extractable material determined as specified in table II.

B = Percent ash determined as specified in table II.

C = Percent hide substance determined as specified in Method 6441 of FED-STD-311.

5. PACKAGING

5.1 Preservation, packing, palletization and marking. The preservation, packing, palletization and marking for each level of protection shall be in accordance with the applicable requirements of MIL-P-43432 (see 6.2).

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The leather is intended for use as upper leather in the Department of Defense service, field, and general purpose footwear.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Treatment required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.3, and 6.3).
e. Color, when black is not required (see 3.6).
f. When the fungicide paranitrophenol is not required (see 3.8).
g. Levels of preservation and packing (see 5.1).
h. When palletization is required (see 5.1).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Sample. For access to samples, address the contracting activity issuing the invitation for bids or request for proposal.

6.5 Selection of hides. It is recommended that hides of 40 to 65 pounds, of not too spready nature, be used for this leather.

6.6 Tanner. It is suggested that the tanner obtain from the supplier of the lubricating materials, a certificate which indicates the percent hydrocarbon mineral oil contained in the lubricating materials.

6.7 Water resistant treatment. Silicone compounds have been found to be capable of imparting the required water resistant properties to the leather.

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6.8 Subject term (key word) listing.

Boot
Grain
Hide
Shoe

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - NU
Air Force - 11

Preparing Activity:

Army - GL

(Project 8330-0188)

Review activities:

Army - MD
Air Force - 82, 99
DLA - CT

User activity:

Navy - MC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-L-3122G	1991 March 29

3. DOCUMENT TITLE

Leather, Cattlehide, For Footwear Uppers, Chrome Tanned, Fatliquored

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
U.S. Army Natick RD&E Center	508-651-4532	256-4532
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-IRT Natick, MA 01760-5019	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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